

PRAVEEN DAS

Full-Stack Developer (MERN)

☎ +918848990353 ✉ praveendask97@gmail.com 🌐 praveen-das-k 📁 portfolio 📍 Kerala, India

PROFESSIONAL SUMMARY

Full-Stack Developer specializing in the **MERN stack**, **TypeScript**, and backend engineering. Builds scalable web applications and APIs with **Node.js**, **Express.js**, **React.js**, **MongoDB**, and **PostgreSQL**. Experienced in **WebSockets**, **Redis**, **Kafka**, authentication and authorization, microservices, event-driven architecture, **AWS**, **Docker**, **CI/CD**, and **Generative AI/RAG**. Focused on reliable backend services, real-time systems, data modeling, and production-oriented application architecture.

EDUCATIONS

St Thomas College | Palai

Bachelor's Degree

- **Sustainable Agriculture**, St Thomas College, Palai

PROJECTS

Docsy – Enterprise Document Intelligence & Real-Time Chat Platform | 🌐 Website Sep 2026 - Present

Personal Project

- Architected a full-stack **RAG document intelligence platform** using Next.js, TypeScript, PostgreSQL, and Drizzle ORM for grounded, multi-turn PDF interactions.
- Engineered an asynchronous document pipeline with Upstash QStash and **LangChain for extraction**, recursive **chunking**, and batch **768D Gemini embeddings**.
- Implemented **Pinecone semantic search** with tenant/document-scoped namespaces, retrieving the top-5 passages for context-aware LLM responses.
- Built durable LLM token streaming with Upstash Workflow and Realtime, enabling **replayable chunks**, **reconnect recovery**.
- Enforced security with Clerk authentication, **cryptographic webhook verification**, daily query quotas, and Zustand-based global state management.

Technologies / Tools Used : Next.js 16, TypeScript, PostgreSQL, Drizzle ORM, Redis, Pinecone, Google Gemini, LangChain, Upstash Workflow, Zustand, Tailwind CSS, Clerk, Stripe

Virel –Social Discovery & Travel Planning Platform | 🌐 Website Jan 2026 - Present

Personal Project

- Architected a **location-based platform** using **React**, **TypeScript**, **Node.js**, **Express**, and **PostgreSQL/Prisma**, encompassing **15+ models** and **15+ REST API modules**.
- Engineered a high-throughput feed with **Redis Sorted Sets** and pipelined MGET caching, serving timelines from memory to eliminate database read bottlenecks.
- Built geospatial clustering using **PostGIS** bounding boxes, Supercluster, and Redis tile caching, dynamically aggregating locations across map zoom levels.
- Developed regional trending with gravity time-decay scoring and **4 BullMQ worker pipelines** for asynchronous indexing and scheduled event processing.
- Containerized the backend with multi-stage Docker and automated CI/CD using **GitHub Actions**, **AWS ECR**, and **EC2**, supported by Vitest test coverage.

Technologies / Tools Used : React.js, Node.js, Express.js, PostgreSQL, Postgis, OLAMaps, Redis, GPS Navigation, Tailwind CSS

Chatvia –Real-Time Chat Application | 🌐 Website Jun 2024 - Present

Personal Project

- Architected a distributed chat platform using **Turborepo**, **Next.js**, **Node.js**, **TypeScript**, **MongoDB**, and **Docker**, comprising **5 dedicated microservices**.
- Built an event-driven messaging pipeline with **Apache Kafka** and **Redis Pub/Sub**, batching consumers at **64KB/1.5s** to reduce MongoDB write saturation.
- Implemented client-side **E2EE** using Web Crypto API with **ECDH P-256**, **AES-GCM-256**, and **100K-iteration PBKDF2**, including automated key rotation.
- Designed offline-first synchronization with **Zustand**, **IndexedDB**, and **Redis**, using version/timestamp deltas to minimize conversation and receipt sync payloads.
- Integrated streaming AI with **Vercel AI SDK** and **Gemini 2.5 Flash**, utilizing Redis-resumable streams and **virtua** virtualization for **60 FPS rendering**.

Technologies / Tools Used : React.js, Node.js, Express.js, MongoDB, Redis, WebSockets, Kafka, Tailwind CSS

SKILLS

Languages/Frameworks :	JavaScript, TypeScript, React.js, Next.js, Node.js, Express.js
Database/Cloud :	MongoDB, PostgreSQL, Redis, Valkey, Prisma, AWS, Firebase
Others :	REST APIs, WebSockets, Message Queues, Docker, CI/CD, Git, Tailwind CSS, Redux, Three.js